PromoLens: Production-Grade AI Microdrama Promo Extraction & Ranking

Target Track: Problem 1 — Find the Promos (Ads) | Evaluation Dataset: 50 Screenplay Episodes (Season 1)

1. Problem Choice, Strategic Framing & Core Assumptions

Why Problem 1? At Kuku, AI microdramas are acquired and monetized primarily through short vertical video ads on TikTok, Instagram Reels, and YouTube Shorts. Creative editing is today's primary operational bottleneck: human editors spend ~18 hours per series scrubbing 50+ episodes to find 20–60s hooks. Problem 1 directly impacts customer acquisition cost (CAC), viewer retention, and content velocity. It provides an objective, benchmarkable challenge where narrative psychology meets ad performance.

Core Operational Assumptions:

- **Strict Platform Duration (20s–60s):** <20s lacks narrative substance to establish conflict; >60s fails vertical ad platform guidelines.
- **Zero-Splice Boundary Alignment:** Cuts must strictly snap to natural dialogue utterance onsets and breath/punctuation terminations or audio stingers. Never cut an actor mid-word.
- **3-Second Hook Rule:** Cold TikTok scrolling requires immediate high entropy in the first 3s (visceral betrayal, public humiliation, slap, or status threat).
- **Zero-Lore Standalone Clarity:** A cold viewer at 1 AM must instantly identify who is wronged and what is at stake without backstory.
- **Zeigarnik Cliffhanger:** Abrupt cut precisely at the peak of unresolved emotional tension, compelling immediate app download.

2. System Architecture: Why a Hybrid Neuro-Symbolic Pipeline?

Brute-forcing 50 full scripts (~150,000 words) into an LLM produces hallucinated timestamps, broken sentence boundaries, high latency, and prohibitive API bills (~$8.50/show). Pure keyword regex lacks semantic understanding of unresolved cliffhangers. We built **PromoLens** as a modular, 5-stage hybrid architecture:

- **Stage 1: Deterministic Screenplay Parser:** Robust state machine parsing 50 episodes, extracting 100 scenes, 3,727 camera/action elements, 1,487 dialogue lines with parenthetical emotional cues (`Cold, Commanding`), and normalizing micro-timestamps (`MM:SS.ss`).
- **Stage 2: Constraint-Solver Window Generator:** Slides boundary-aligned temporal windows adhering strictly to [20.0s, 60.0s], evaluating 32,984 candidate permutations across the season in <1.1 seconds.
- **Stage 3: Multi-Dimensional Scoring Engine:** Evaluates each window via weighted vector scoring:
  Composite Score = 0.35 · Hook + 0.30 · Cliffhanger + 0.20 · Standalone Clarity + 0.15 · Pacing
- **Stage 4: Temporal Non-Maximum Suppression (NMS):** Implements Intersection over Union (IoU ≥ 0.30) to eliminate redundant overlapping cuts of identical scenes, ensuring season-wide promo diversity.
- **Stage 5: Swappable Model Bridge:** Runs 100% offline via our calibrated heuristic scorer, while exposing an swappable interface for Gemini 1.5 Flash / GPT-4o-mini if an interviewer provides an API key.

3. Real-World Scale, Unit Economics & Human-in-the-Loop

Evaluation Dimension	Brute-Force LLM Approach	PromoLens Hybrid Engine
Execution Time (50 Episodes)	~12 to 15 minutes	**1.07 seconds (0.02s / episode)**
Compute / API Cost per Show	~$8.50 per 50-ep series	**$0.00 (Local) / $0.08 (Pruned LLM)**
Monthly Spend (100 Shows)	~$850.00 / month	**<$8.00 / month (>99% reduction)**
Boundary Alignment Precision	68% (frequent mid-word cuts)	**100% Exact Math Alignment**
Production Asset Output	Raw unstructured text	**JSON, CSV, and Premiere Pro EDLs**

Human-in-the-Loop (HITL) Workflow: Video editors never scrub blind. PromoLens generates instant CMX 3600 Edit Decision Lists (EDLs) imported directly into Adobe Premiere Pro or DaVinci Resolve. Editors review only the top-ranked candidates, reducing promo delivery time from 18 hours to under 45 minutes per show.

4. Strategic Ranking: Which Promos to Post First & Why?

Ad creative performance depends entirely on platform audience psychology and ad fatigue cycling. Promos are ranked into 3 tactical funnels:

Posting Wave & Funnel	Top Extracted Promo & Timestamps	Why Post First? (Psychological Rationale)
Wave 1: Cold User Acquisition (Top-of-Funnel / Post FIRST) *TikTok & Meta Broad Reels*	**Rank #1: Ep 2 [00:53.40 – 01:16.40] (23.0s)** **Hook:** *'You look like a kitchen maid who wandered into the wrong wedding.'* **Cliffhanger:** Juliana turns with fury: *'Dignity?'* **Rank #2: Ep 2 [00:00.00 – 00:27.40] (27.4s)** **Hook:** Gerald dumps bride at altar: *'I cannot marry you.'*	Cold scrolling viewers have **zero affinity with lore**. Only high-contrast moral outrage, public humiliation, and cruel betrayal trigger thumb-stops in <2s. High completion rate without prior exposition.
Wave 2: Emotional Escalation (Mid-Funnel / Post SECOND) *Meta Video & In-App Ads*	**Rank #3: Ep 30 [01:14.80 – 01:48.80] (34.0s)** **Hook:** *'Give those facts to a magistrate in order, and she becomes a poisoner.'* **Cliffhanger:** Catherine demands: *'And the Dowager?'* **Rank #5: Ep 26 [00:55.80 – 01:35.00] (39.2s)** **Hook:** *'Rupert & Catherine are poisoning me in bed.'*	Retargets viewers who engaged with Wave 1. Introduces life-or-death stakes, murder framing, and protective romance. Drives consideration and series bingeing into direct app installs.
Wave 3: Cathartic Vindication (Retargeting / Post THIRD) *YouTube Shorts & Coin Unlocks*	**Episode 42 [00:17.90 – 00:54.00] (36.1s)** **Hook:** Corrupt lord yells: *'Guards! Arrest this imposter!'* **Cliffhanger:** The broken soldier emerges in full ducal regalia, and royal guards salute him.	Re-activates stalled or churned users. Delivers the ultimate revenge payoff (the broken beggar is secretly the Grand Duke), compelling users to unlock paid episodes.

5. Quality Checking, Error Auditing & Honest Failure Analysis

We ran an automated diagnostic test suite (`src/quality_checker.py`) across all generated promos. Of 25 top promos, **17 passed clean for immediate render**, 3 were flagged for context review, and 5 required cut boundary tuning. Here is where the tool picks badly and our mitigations:

- **1. Pronoun Antecedent Ambiguity (Cold-Viewer Confusion):**
  Failure Mode: In Ep 32 [00:47.80–01:19.40], opening line is: *'A disgraced woman? Your soldier?'* Without prior context, cold viewers do not know who 'your soldier' refers to.
  Remedy: Built `PRONOUN_AMBIGUITY` detector that flags cuts opening on unresolved 3rd-person pronouns, routing them to editorial review.
- **2. Non-Verbal Tension Blind Spot (Visual Action vs Low Text Density):**
  Failure Mode: In Ep 1 [02:11.30–02:17.00], Juliana's knuckles whiten and she refuses to weep as dark strings swell. Pure NLP models score this low due to zero dialogue.
  Remedy: Weighted `SOUND FX` ('cliffhanger drone swells') and `ACTION` ('knuckles whiten') to reward non-verbal visual acting.
- **3. False Cliffhangers (Cut on Declarative Statements):**
  Failure Mode: Dialogue ending on flat periods (e.g., *'You have a solution.'*) sounds like an ordinary pause rather than an unresolved cliffhanger.
  Remedy: Implemented interrogative boost (`?`), penalizing non-interrogative endings unless paired with sharp audio stingers or physical climaxes.
- **Tool Reflection (Where AI Failed Us):** Early zero-shot prompts to LLMs produced hallucinated cutoffs (e.g. cutting Juliana off at 'aren't t'). We completely eliminated this flaw by constraining candidate generation to deterministic script element boundaries.

6. Deliverables & Prototype Verification

- **Live Cloud Deployed App:** **https://integenpiyush-kukufm-promolens-app-m1j9gq.streamlit.app/**
- **GitHub Repository & Code:** https://github.com/Integenpiyush/kukufm-promolens
- **One-Command Production CLI:** `python run_pipeline.py` — Evaluates all 50 episodes in 1.07s; exports JSON, CSV, and Premiere EDLs.
- **Raw Outputs Folder:** Full unedited results in `raw_outputs/` (includes `all_promos.json`, `top_ranked_promos.csv`, and 10 NLE EDLs).